

Strategic Practice 4

Distributions and E.V.s for discrete r.v.s

1) 2 discrete r.v.s X & Y on same sample space

yes, it's possible X and Y have identical distributions but $X=Y$ never occurs

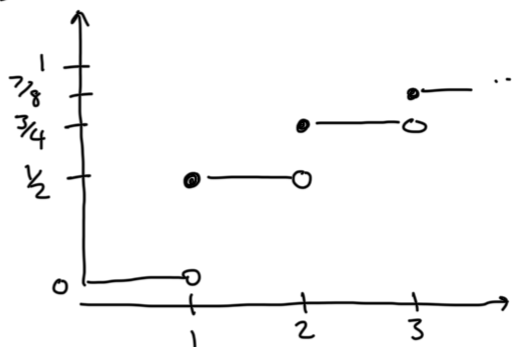
example: X : indicator of flipping H $\left. \begin{array}{l} Y: \text{ " " " } \end{array} \right\} \text{ fair coin}$

2) X be random day of week
and takes values $1, \dots, 7$ w/ equal prob.
1 - Mon, 2 - Tuesday, etc.

Let Y be the next day after X represented by an integer $1, \dots, 7$

X & $Y \rightarrow$ have same distribution
 $P(X < Y) = 6/7$

3) $X - 1 \sim \text{Geom}(\frac{1}{2})$
 $P(X = k) = (\frac{1}{2})^k$



correction
question asks for CDF

$$\therefore P(X \leq x) = 1 - (\frac{1}{2})^{L(x)}$$

4) Yes, it's possible $E(X) > 100$
but $Y > X$ w/ probability > 0.99

$$E(Y) = 1 \cdot 1.000$$

5) X is a discrete r.v. w/ possible values $1, 2, 3, \dots$
Let $F(x) = P(X \leq x)$ be the CDF of X

show $E(X) = \sum_{n=0}^{\infty} (1 - F(n))$

$$\begin{aligned}
 \downarrow \\
 E(X) &= \sum_{x=0}^{\infty} x P(X=x) \\
 &= 1 - F(0) \\
 &\quad + 1 - F(1) \\
 &\quad + 1 - F(2) \\
 &\quad + \dots \\
 &= 1 - P(X \leq 0) \\
 &\quad + 1 - P(X \leq 1) \\
 &\quad + 1 - P(X \leq 2) \\
 &\quad + \dots \\
 &= P(X > 0) \\
 &\quad + P(X > 1) \\
 &\quad + P(X > 2) \\
 &\quad + \dots \\
 &= P(X=1) + P(X=2) + P(X=3) \dots \\
 &\quad + P(X=2) + P(X=3) \dots \\
 &\quad \quad + P(X=3) \dots \\
 &\quad \quad \quad + \dots \\
 &= \sum_{x=0}^{\infty} x P(X=x)
 \end{aligned}$$

Assume no limit to # of candidates

that for any n candidates the rankings

are equally likely
that there are no ties in rankings

Let X be index of first candidate who ranks
better than C_1

$$E(X) = \underbrace{2}_{C_2} \cdot \frac{1}{2} + 3 \cdot \frac{1}{3!} + 4 \cdot \frac{2!}{4!} + 5 \cdot \frac{3!}{5!} + \dots$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = \infty$$

Note from answer key: Just as in the
St Petersburg paradox
there must be in reality an
upper bound on the # of
rounds

Indicator r.v.s & Linearity of Expectation

- 1) 50 people comparing birthdays
Assume birthdays are independent

$$E(\# \text{ of pairs}) = \binom{50}{2} \frac{1}{365} \approx 3.356 \quad \text{by linearity}$$

$E(\# \text{ of days with at least 2 people born})$

$$= 365 \cdot \left(1 - \underbrace{\left(\frac{364}{365} \right)^{50}}_{\substack{\uparrow \\ \text{no one born on} \\ \text{specified day}}} - 50 \cdot \underbrace{\frac{1}{365}}_{\substack{\uparrow \\ \text{exactly one person} \\ \text{born on specified day}}} \cdot \left(\frac{364}{365} \right)^{49} \right)$$

- 2) 20 bags of gummies $\rightarrow$ randomly distributed to
20 students

(outcome of each bag
distribution is independent)

$E(\# \text{ of bags first 3 students get})$

X_i is r.v. for # of bags student i gets
 $i \in \{1, \dots, 20\}$

$$X_i \sim \text{Bin}(20, \frac{1}{20})$$

$$E(X_1 + X_2 + X_3) = E(X_1) + E(X_2) + E(X_3) = 3$$

$E(\# \text{ of students who get at least one bag})$

$$Y_i \rightarrow \begin{cases} 1 & \text{when } X_i \geq 1 \\ 0 & \text{when } X_i = 0 \end{cases}$$

$$E(Y_1 + \dots + Y_{20}) = 20 E(Y_1) = 20 \left(1 - \left(\frac{19}{20}\right)^{20}\right)$$

by symmetry

3) 100 shuffles: at each stage you tie two randomly selected ends together. This results in either a loop or longer lace

$E(\# \text{ of stages until everything is looped}) = 100$

(story explanation: 200 loose ends & 2 ends eliminated at each stage)

$E(\# \text{ of loops after everything is looped})$

X_i is r.v. for each stage i , $i \in \{1, \dots, 100\}$

$$X_i: \begin{cases} 0 \rightarrow \text{longer lace} \\ 1 \rightarrow \text{loop formed} \end{cases}$$

$$\begin{aligned}
 E(\# \text{ of loops}) &= 1 \cdot P(X_1=1) + 1 \cdot P(X_2=1) + \dots \\
 &\quad \frac{100}{\binom{200}{2}} \quad \frac{99}{\binom{198}{2}} \quad \dots \\
 &= \sum_{n=1}^{100} \frac{n}{\binom{2n}{2}} = \sum_{n=1}^{100} \frac{1}{2n-1}
 \end{aligned}$$

- 4) for each name $x \rightarrow$ hash fx_n h is used
 $\rightarrow h(x)$ is location of x 's #
 $\rightarrow h(x)$ is deterministic
 $\rightarrow$ assume h is truly random

Let there be k ppl w/ each person's # stored in a random location (independently) represented by an int. b/w 1 & n

It then might happen that one location has more than one phone # stored

X_i is r.v. $\rightarrow 0$ if something stored in loc. i
 $\rightarrow 1$ if nothing in loc. i

$$\begin{aligned}
 E(\# \text{ of empty locations}) &= E(X_1) + \dots + E(X_n) \\
 &= \left(\frac{n-1}{n}\right)^k \cdot n
 \end{aligned}$$

Y_i is r.v. $\rightarrow 1$ exactly 1 # in loc. i
 $\rightarrow 0$ otherwise

$$\begin{aligned}
 E(\# \text{ of locations w/ exactly 1 #}) &= E(Y_1) + \dots + E(Y_n) \\
 &= n \cdot \frac{k}{n} \left(\frac{n-1}{n}\right)^{k-1}
 \end{aligned}$$

Z_i is r.v. $\rightarrow 1$ multiple #s in loc. i

○ otherwise

$$\begin{aligned} E(\# \text{ of locations w/ multiple \#s}) &= E(Z_1) + \dots + E(Z_n) \\ &= n \cdot \left(1 - \left(\frac{n-1}{n} \right)^k - \frac{k}{n} \left(\frac{n-1}{n} \right)^{k-1} \right) \end{aligned}$$